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1) $p(\text{action}) = \frac{3}{5}$ $p(\text{comedy}) = \frac{2}{5}$, $\frac{\text{count}(w_i, c) + 1}{(\sum_{w \in V} \text{count}(w, c)) + |V|}$ so,

$p(\text{fast} \mid \text{comedy}) = \frac{1+1}{9+7} = \frac{2}{16}$

$p(\text{couple} \mid \text{comedy}) = \frac{2+1}{9+7} = \frac{3}{16}$

$p(\text{shoot} \mid \text{comedy}) = \frac{0+1}{9+7} = \frac{1}{16}$

$p(\text{fly} \mid \text{comedy}) = (1+1)/(9+7) = 2/16$

$p(\text{fast} \mid \text{action}) = \frac{2+1}{11+7} = 3/18$

$p(\text{couple} \mid \text{action}) = \frac{0+1}{11+7} = 1/18$

$p(\text{shoot} \mid \text{action}) = \frac{4+1}{11+7} = \frac{5}{18}$

$p(\text{fly} + \text{action}) = \frac{1+1}{11+7} = \frac{2}{18}$

$p(\text{comedy} \mid D) = \left(\frac{2}{16} \cdot \frac{3}{16} \cdot \frac{1}{16} \cdot \frac{2}{16}\right) \cdot \left(\frac{2}{5}\right) = 0.00007324218$

$p(\text{action} \mid D) = \left(\frac{3}{18} \cdot \frac{1}{18} \cdot \frac{5}{18} \cdot \frac{2}{18}\right) \cdot \left(\frac{3}{5}\right) = 0.00017146776$

Because $0.00017 > 0.000073$, sentence D belongs to $\boxed{\text{"action"}}$.

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Euclidean distance $d(c, p) = \sqrt{(c_x - p_x)^2 + (c_y - p_y)^2}$ s.t.

Iteration 1:

2) cluster 1: $\boxed{gene_1 = (-4-(-3))^2 + (-3-0)^2 = 10}$

$gene_2 = (6-(-3))^2 + (5-0) = 106$

$gene_3 = (1-(-3))^2 + (-7-0)^2 = 65$

$\boxed{gene_4 = (-4-(-3))^2 + (-6-0)^2 = 37}$

$gene_5 = (4-(-3))^2 + (6-0)^2 = 85$

$\boxed{gene_6 = (-1-(-3))^2 + (5-0)^2 = 29}$

$\boxed{gene_7 = 0}$

cluster 1: gene1, gene4, gene6, gene7

cluster 2: $gene_1 = (-4-3)^2 + (-3-0)^2 = 58$

$\boxed{gene_2 = (6-3)^2 + (5-0)^2 = 34}$

$\boxed{gene_3 = (1-3)^2 + (-7-0)^2 = 53}$

$gene_4 = (-4-3)^2 + (-6-0)^2 = 85$

$\boxed{gene_5 = (4-3)^2 + (6-0)^2 = 37}$

$gene_6 = (-1-3)^2 + (5-0)^2 = 41$

$gene_7 = (-3-3)^2 + (0-0)^2 = 36$

$\boxed{gene_8 = 0}$

cluster 2: gene2, gene3, gene5, gene8

New cluster centroids: $c = \frac{1}{p} \sum_{p \in c_i} p$ s.t.

$(x, y) = \left(\frac{-4-4-1-3}{4}, \frac{-3-6+5+0}{4}\right) = \boxed{(-3, -1)}$ for cluster 1

$(x, y) = \left(\frac{6+1+4+3}{4}, \frac{5-7+6+0}{4}\right) = \boxed{(3.5, 1)}$ for cluster 2

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2 continued)

Iteration 2:

Cluster 1:

$\boxed{gene1 = (-4+3)^2 + (-3+1)^2 = 5}$

$gene2 = (6+3)^2 + (5+1)^2 = 117$

$\boxed{gene3 = (1+3)^2 + (-7+1)^2 = 52}$

$\boxed{gene4 = (-4+3)^2 + (-6+1)^2 = 26}$

$gene5 = (4+3)^2 + (6+1)^2 = 98$

$gene6 = (-1+3)^2 + (5+1)^2 = 40$

$\boxed{gene7 = (-3+3)^2 + (0+1)^2 = 1}$

$gene8 = (3+3)^2 + (0+1)^2 = 37$

Cluster 2:

$gene1 = (-4-3.5)^2 + (-3-1)^2 = 72.25$

$\boxed{gene2 = (6-3.5)^2 + (5-1)^2 = 21}$

$gene3 = (1-3.5)^2 + (-7-1)^2 = 69$

$gene4 = (-4-3.5)^2 + (-6-1)^2 = 105.25$

$\boxed{gene5 = (4-3.5)^2 + (6-1)^2 = 25.25}$

$\boxed{gene6 = (-1-3.5)^2 + (5-1)^2 = 36.25}$

$gene7 = (-3-3.5)^2 + (0-1)^2 = 43.25$

$gene8 = (3-3.5)^2 + (0-1)^2 = 1.25$

cluster 1 = gene1, gene3, gene4, gene7 cluster 2 = gene2, gene5, gene6, gene8

$$(x,y) = \left(\frac{-4+1-4+3}{4}, \frac{-3-7-6+0}{4}\right) = \boxed{(-2.5, -4)}$$

for cluster 1

$$(x,y) = \left(\frac{6+4-1+3}{4}, \frac{5+6-5+0}{4}\right) = \boxed{(3,4)}$$

for cluster 2

Iteration 3:

$\boxed{gene1 = (-4+2.5)^2 + (-3+4)^2 = 3.25}$

$gene2 = (6+2.5)^2 + (5+4)^2 = 153.25$

$\boxed{gene3 = (1+2.5)^2 + (-7+4)^2 = 21.25}$

$\boxed{gene4 = (-4+2.5)^2 + (-6+4)^2 = 7}$

$gene5 = (4+2.5)^2 + (6+4)^2 = 142.25$

$gene6 = (-1+2.5)^2 + (5+4)^2 = 83.25$

$\boxed{gene7 = (-3+2.5)^2 + (0+4)^2 = 16.25}$

$gene8 = (3+2.5)^2 + (0+4)^2 = 46.25$

$gene1 = (-4-3)^2 + (-3-4)^2 = 98$

$\boxed{gene2 = (6-3)^2 + (5-4)^2 = 10}$

$gene3 = (1-3)^2 + (-7-4)^2 = 125$

$gene4 = (-4-3)^2 + (6-4)^2 = 149$

$\boxed{gene5 = (4-3)^2 + (6-4)^2 = 5}$

$\boxed{gene6 = (-1-3)^2 + (5-4)^2 = 17}$

$gene7 = (-3-3)^2 + (0-4)^2 = 52$

$\boxed{gene8 = (3-3)^2 + (0-4)^2 = 16}$

Clusters <u>stay same</u> because of this we stop.

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3) $\beta_1 = 0$, $\beta_2 = 0$, L (learning rate) $= 1$

$\beta_1' = -\frac{2}{n}\sum_{i=0}^{n} x_i(y_i - \bar{y}_i)$, $\sum_{i=0}^{n} x_i(y_i - \bar{y}_i) = \frac{\partial}{\partial \theta_0} J(\theta_0, \theta_1)$

Iteration 1:

$\beta_2' = -\frac{2}{n}\sum_{i=0}^{n}(y_i - \bar{y}_i)$, $\beta_1' = -\frac{1}{2}(1(12-0)+2(15-0)+3(10-0)+4(2-0)) = -42.5$

$\beta_2' = -\frac{1}{2}((12-0)+(15-0)+(10-0)+(2-0)) = -22$

$\beta_1 = \beta_1 - 1 * \beta_1'$, $\beta_2 = \beta_2 - 1 * \beta_2'$

$\beta_1 = 0 - 1 * (-42.5) = \boxed{42.5}$, $\beta_2 = 0 - 1 * (-22) = \boxed{22}$

Iteration 2:

$\beta_1' = -\frac{1}{2}(1(12-(42.5)) + 2(15-(42.5)) + 3(10-(42.5)) + 4(2-42.5))$
$= 120$

$\beta_2' = -\frac{1}{2}(1(12-(22)) + 2(15-(22)) + 3(10-(22)) + 4(2-(22)))$
$= \boxed{-132}$

$\beta_1 = 42.5 - 1 * (120) = \boxed{-122.5}$, $\beta_2 = 22 - 1 * (-132) = \boxed{159}$

Iteration 3: $\beta_1' = -\frac{1}{2}(1(12-122.5) + 2(15-122.5) + 3(10-122.5) + 4(2-122.5)) =$
$= 534,25$

$\beta_2' = -\frac{1}{2}(1(12-159) + 2(15-159) + 3(10-159) + 4(2-159)) = -252.5$

$\beta_1 = 122.5 - 1 * (534,25) = -462.25$, $\beta_2 = 159 - 1 * (-252.5) = -593$